

# SMART INDIA HACKATHON 2026

## TITLE PAGE

**PROBLEM STATEMENT ID**

**26189**

**PROBLEM STATEMENT**

**AI POWERED CRIMINAL  
NETWORK ANALYSIS**

**THEME**

**POWERED AI FOR  
ACCLERATING  
INVESTIGATION**

**PS CATEGORY**

**SOFTWARE**

**TEAM ID**

**TEAM NAME**

**CYBER RAKSHAK**



# AI-Powered Criminal Network Analysis & Investigation Portal

## What we are introducing?

**An AI-driven platform that combines heterogeneous public and investigative data to identify suspicious patterns, analyze relationships between entities, and assist investigators in discovering potential criminal networks.**

## Key Innovation

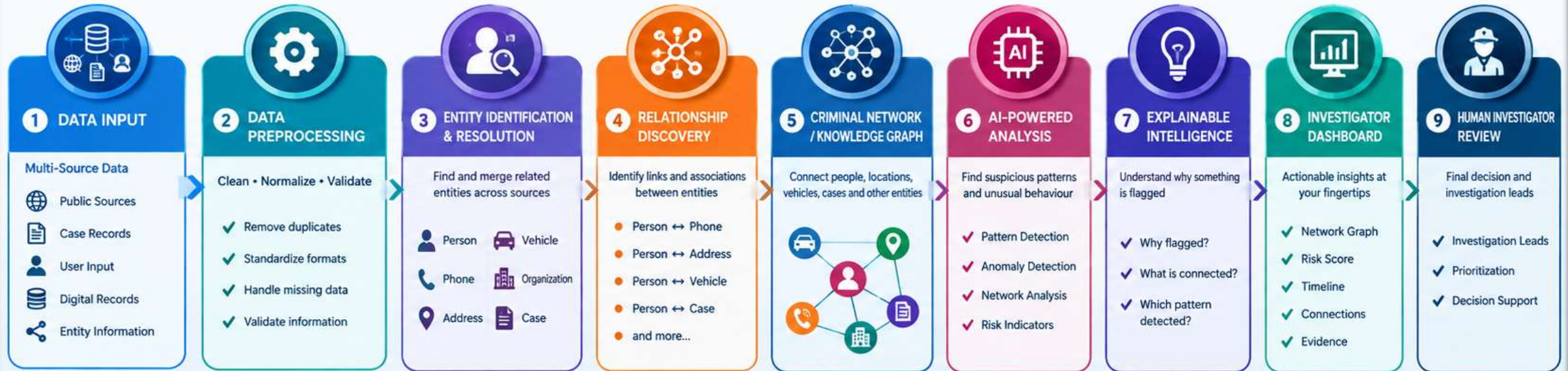
| Feature                       | Purpose                                                     |
|-------------------------------|-------------------------------------------------------------|
| Multi-Source Data Integration | Combines fragmented investigation data                      |
| Entity Identification         | Identifies people, phones, vehicles, locations, cases, etc. |
| Relationship Mapping          | Finds connections between entities                          |
| Interactive Network Graph     | Visualizes the complete network                             |
| AI Pattern Analysis           | Detects meaningful patterns                                 |
| Anomaly Detection             | Highlights unusual relationships/activities                 |
| Risk Indicator                | Helps prioritize investigation                              |
| Timeline Analysis             | Understands events chronologically                          |
| Case Linking                  | Connects related cases/entities                             |
| Explainable Results           | Shows reasons behind generated indicators                   |
| Investigator Dashboard        | Provides one unified investigation interface                |
| Human-in-the-Loop             | Keeps final judgment with the investigator                  |



## CYBER RAKSHAK

AI-Powered Criminal Network Analysis & Investigation Support System

Connect the data.  
Detect the pattern.  
Reveal the network.



CORE FLOW



COLLECT



CONNECT



ANALYZE



EXPLAIN



INVESTIGATE



AI identifies suspicious patterns and relationships — the final investigative decision remains with the human investigator.

“CYBER RAKSHAK transforms fragmented data into actionable, explainable criminal-network intelligence.”

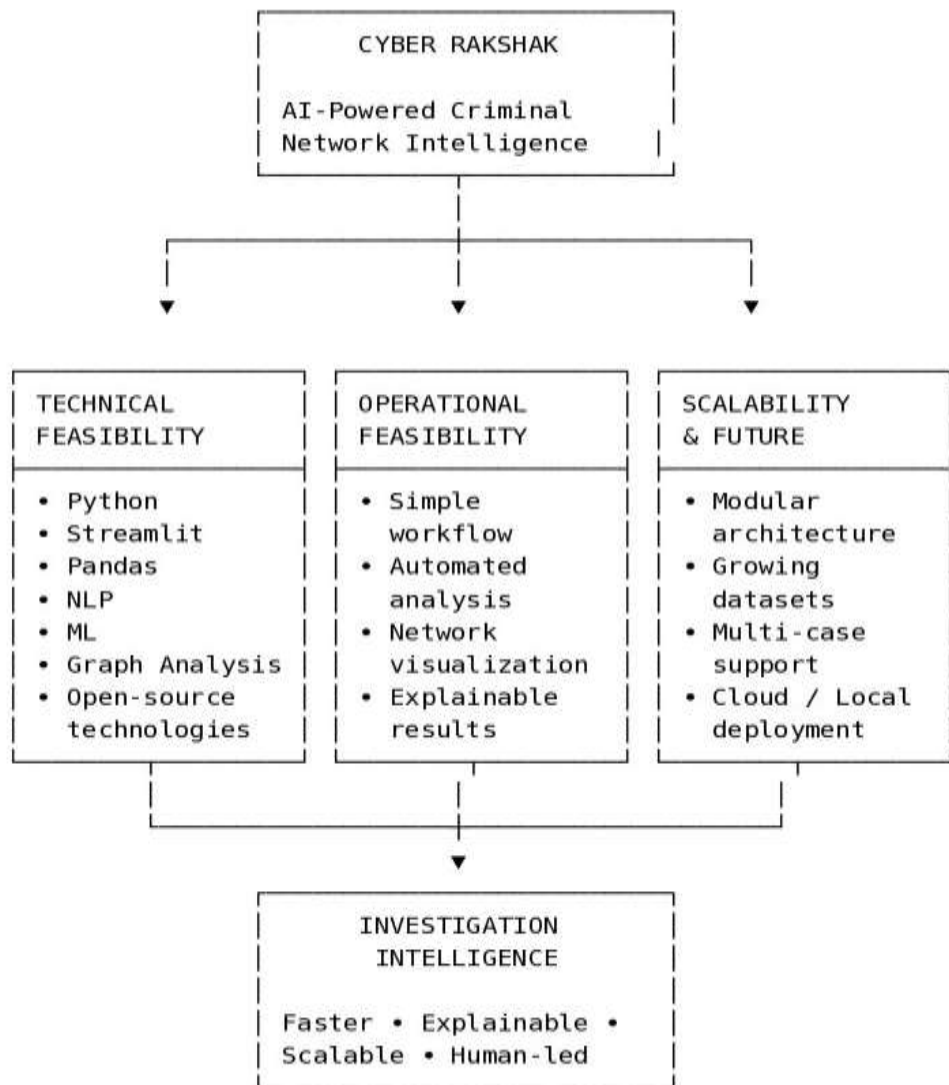
# TECHNICAL APPROACH



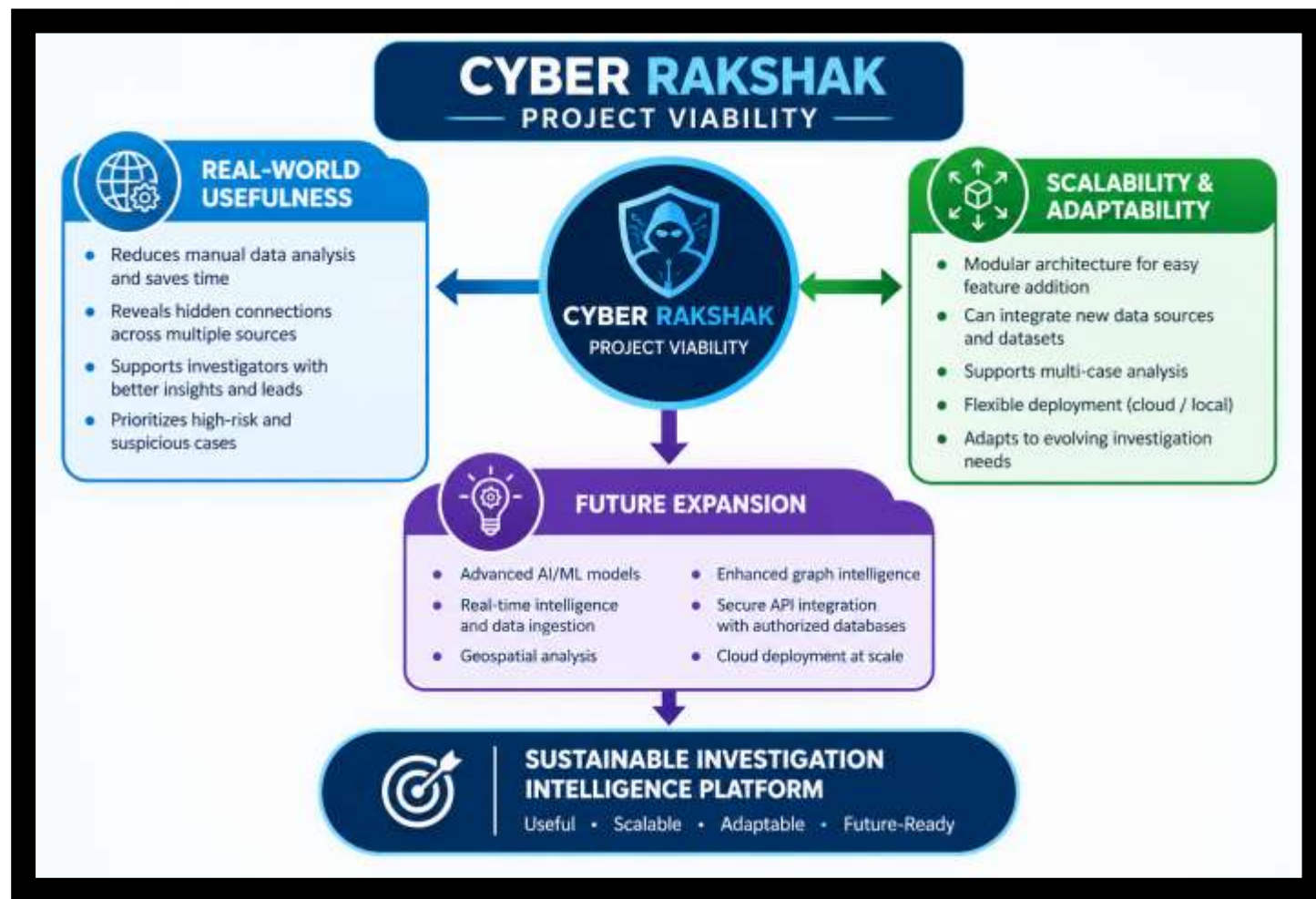
| Innovation               | Technology Used                | Purpose                                                                |
|--------------------------|--------------------------------|------------------------------------------------------------------------|
| Multi-Source Data Fusion | Python, Pandas                 | Collect, clean & combine heterogeneous data                            |
| Entity Resolution        | Python, Fuzzy Matching / NLP   | Identify matching or related entities                                  |
| Relationship Discovery   | NetworkX                       | Find relationships between people, phones, addresses, vehicles & cases |
| Criminal Network Graph   | NetworkX + Graph Visualization | Visualize hidden connections and network structures                    |
| AI Pattern Analysis      | Scikit-learn / ML Models       | Detect anomalies and suspicious patterns                               |
| Text Intelligence        | NLP                            | Extract useful entities and information from text                      |
| Risk Intelligence        | Python + ML/Rule-based Scoring | Generate explainable risk indicators                                   |
| Interactive Dashboard    | Streamlit                      | Provide investigator-facing analysis interface                         |
| Data Storage             | JSON / CSV / Server Storage    | Persist investigation and application data                             |
| Explainable Results      | Rule/Feature-based Explanation | Show why an entity or network was flagged                              |

# FEASIBILITY AND VIABILITY

## FEASIBILITY



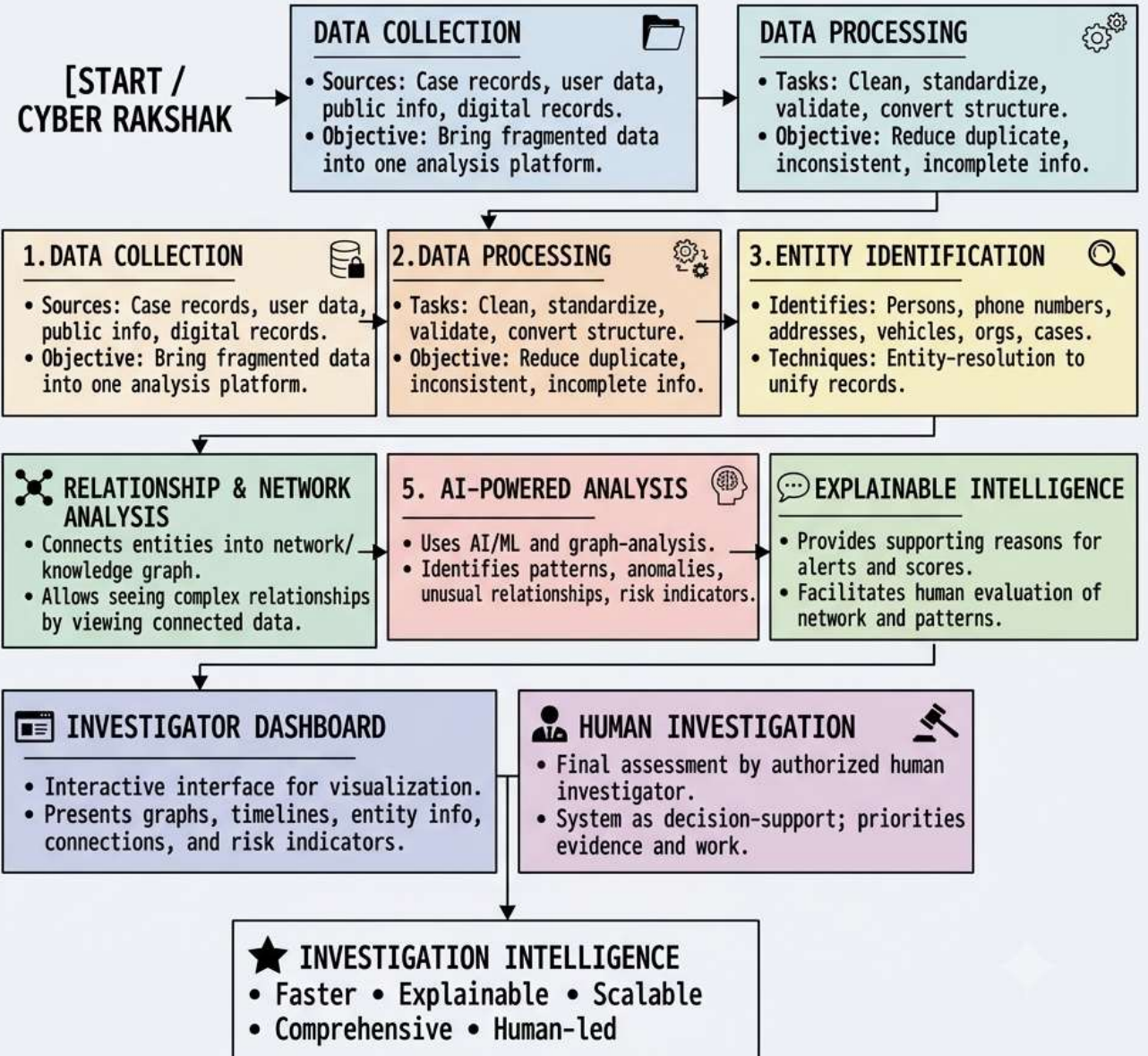
## VIABILITY



# WORKING PROCEDURE ON PORTAL



## CYBER RAKSHAK: AI-POWERED CRIMINAL NETWORK INTELLIGENCE



# BENEFITS



# RESEARCH AND REFERENCES

## BENEFITS OF CYBER RAKSHAK

- ⚡ **Faster Investigation** — Reduces time spent on manual data correlation.
- ↔ **Hidden Connection Discovery** — Reveals relationships between people, cases, locations, vehicles and other entities.
- **AI-Assisted Analysis** — Detects suspicious patterns and anomalies across connected data.
- 📊 **Better Decision Support** — Provides network graphs, timelines, risk indicators and explainable insights.
- 🎯 **Investigation Prioritization** — Helps investigators identify cases and relationships that may require closer attention.
- 🔍 **Centralized Intelligence** — Brings fragmented information into a unified investigation view.
- 👮 **Human-in-the-Loop** — Supports investigators while keeping final decisions with authorized human experts.
- 📈 **Scalable Platform** — Can be expanded with new datasets, analytical models and authorized data sources.

- **USING CHATGPT AND GEMINI:-**
- [https://chatgpt.com/features/deep-research/?utm\\_source=chatgpt.com](https://chatgpt.com/features/deep-research/?utm_source=chatgpt.com)
- **ON YOUTUBE:-**
- [https://youtu.be/qXS75gvT8\\_A?si=0ISiUE2F8FNZtB7c](https://youtu.be/qXS75gvT8_A?si=0ISiUE2F8FNZtB7c)
- [https://youtu.be/qXS75gvT8\\_A?si=0ISiUE2F8FNZtB7c](https://youtu.be/qXS75gvT8_A?si=0ISiUE2F8FNZtB7c)
- **GITHUB:-**
- <https://github.com/iamhars/hsharma108-gif/SIH26189.git>